



Karnataka State Police
Government of Karnataka

Powered by **HACK2SKILL**

Dataathon 2026

Technology partner



Team Details

- a. Team name: My internet is Cooked
- b. Team leader name: Bhakti Chand Tak
- c. Team size: 4
- d. Problem Statement: AI-Driven Crime Analytics & Visualization Platform



A verified data-to-dashboard pipeline for next-week crime intelligence

An end-to-end, leakage-aware AI platform that converts raw police FIR data into 1 week predictive crime intelligence, district-calibrated risk zones, repeat-offender tracking, cyber networks, and real-time scenario simulations all packaged in one portable, zero dependency dashboard.

80,000

Synthetic FIR Records
Grounded in Karnataka

164

Police Stations across
31 Districts

1 week

Chronological
Forecast Horizon

1 HTML

Portable, Embedded HTML
Dashboard

Bullet Points

- Dual-Engine Predictive Intelligence (Count & Serious Risk models)
- District-Calibrated Hotspots (PAI + Voronoi boundaries)
- Cyber & Financial Route Tracking (Repeat accused network graphs)
- Socio-economic crime correlation (strict statistical governance)
- Dynamic Scenario Simulator (Holidays, hours, day-types)
- Trend alerts & anomaly detection



OPPORTUNITIES & SOLUTION DIFFERENTIATION

01

A. HOW DIFFERENT IS IT?

- Unified 10-phase pipeline
- Zero future leakage chronological split
- Live "What-If" scenario simulations

Different - Forecasting, spatial clusters, near-repeat context, fairness diagnostics and cyber routes are delivered together not as disconnected notebooks.

Useful - Station and district views translate model outputs into map-based operational awareness with transparent thresholds and confidence context.

Defensible - Chronological validation, machine-readable sanity reports and explicit synthetic-data limits make every output reviewable

02

B. HOW DOES IT SOLVE PROBLEM?

- Actionable Station Classifications
- Cross-Jurisdiction Cyber Crime Routing
- Empirical Bayes Rate Stabilization

UNIQUE SELLING PROPOSITION (USP)

USP: Verified synthetic engineering + calibrated multi-head forecasting + bipartite cyber route tracking in a single deployable web artifact.



The feature set spans prediction, spatial context, cyber intelligence and auditability

■ Next-Week Crime Forecast

Poisson gradient boosting with **chronological** evaluation.

■ Serious Offence Probability

Calibrated classifier with a validation selected threshold.

■ District PAI (Predictive Accuracy Index) Risk Calibration

Prospective count thresholds selected by district.

■ DBSCAN (Trend Alerts) & Near-Repeat Contagion

Recent DBSCAN clusters plus permutation-tested proximity signals (Haversine clusters + Knox decay).

■ Interactive Scenario Simulator

Month, hour, holiday, day type and category controls.

■ Route-Capable Cyber Graph

Stable accused IDs linked to chronological station activity (Repeat offender station routes).

■ Descriptive Fairness Audit

Observed synthetic arrest-rate and attention-zone comparisons and Social economic co-relation **Metro areas** correlate higher with **cyber, financial, and property crimes**,

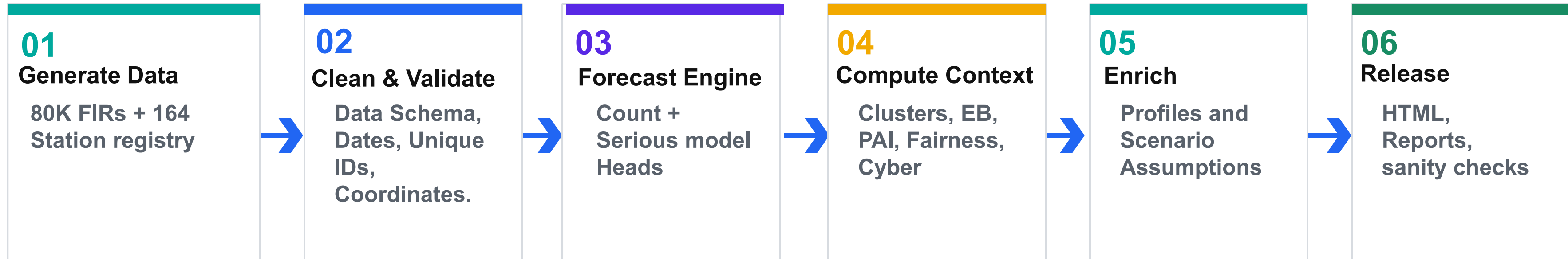
■ Automated Sanity Suite

Automated checks for coordinates, labels, networks and metrics.



The build flow moves from reproducible data to verified dashboard release

6-STAGE PIPELINE SEQUENCE



Note

run_all.bat / run_all.sh to reproduces the complete sequence in one click.



4 - STEP OPERATOR JOURNEY

1

SET SCENARIO CONTEXT

Choose crime profile, holiday, day type, month and hour.

2

SCAN STATEWIDE MAP

Filter 164 Station markers by District and HOTSPOT (WATCH / SAFE label) .

3

INSPECT STATION EVIDENCE

Open Station Forecast, Serious Probability, Uncertainty and Crime mix.

4

TRACE CYBER ROUTES

Focus an accused or station and map chronological Station - level activity.

PROTOTYPE EXPERIENCE

One Interface, Four Evidence Modes

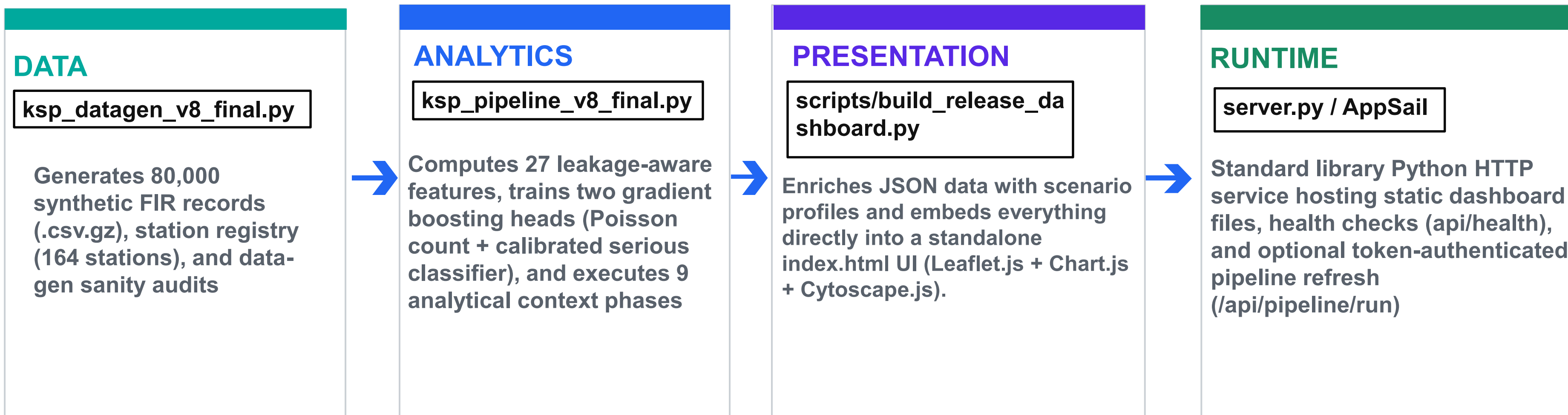
MAP + SIMULATOR • TRENDS • CYBER ROUTES • AUDIT

Our Prototype :
Turns statewide signals into station level context



ARCHITECTURE DIAGRAM OF THE PROPOSED SOLUTION

4-TIER ARCHITECTURE FLOW





The technology stack is deliberately small, open and reproducible

DATA & COMPUTATION

Python 3.11 · NumPy · Pandas · SciPy

MACHINE LEARNING

scikit-learn: Gradient boosting, Calibration, DBSCAN, BallTree

GEOSPATIAL + GRAPH

Shapely · NetworkX · Haversine distance · Station - area approximation

BROWSER EXPERIENCE

HTML · CSS · JavaScript · Leaflet · Chart.js · Cytoscape.js

PACKAGING & DELIVERY

JSON · gzip CSV · Docker · GitHub Actions · Zoho Catalyst AppSail



ZOHO CATALYST SERVICES USED IN THE SOLUTION

PRIMARY CATALYST SERVICE

USED NOW

AppSail

ZOHO CATALYST APPSAIL (USED NOW)

- Docker-based Python microservice (server.py)
- Static Dashboard + REST
- Zero cloud DB dependencies

■ Runtime variables

Configures X_ZOHO_CATALYST_LISTEN_PORT, HOST, ENABLE_PIPELINE_RUN, and ADMIN_TOKEN via Catalyst AppSail console

■ Protected execution

Admin refresh endpoint (/api/pipeline/run) requires ENABLE_PIPELINE_RUN=1 and X-Admin-Token or Bearer authentication

■ Atomic publication

Rebuilds occur under runtime/ before JSON and HTML are replaced.

■ Future Catalyst services

Object storage, scheduled jobs and managed data services are not required by V8.

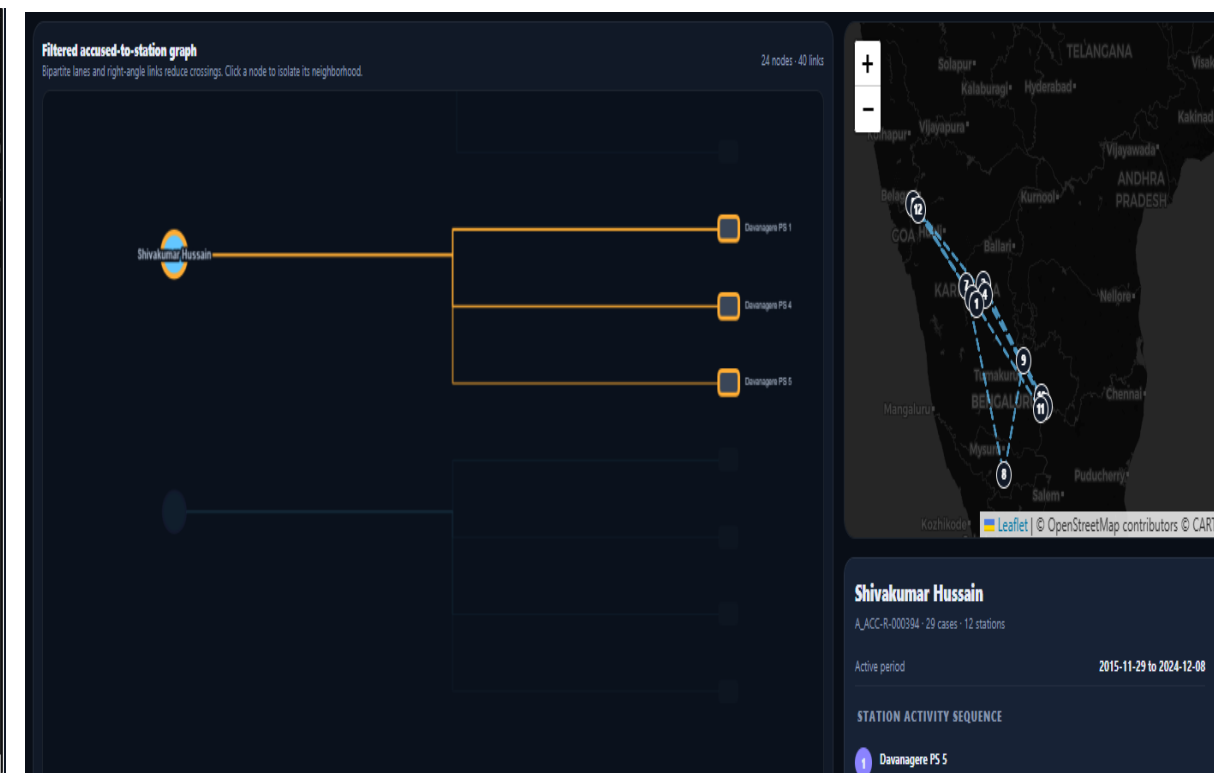
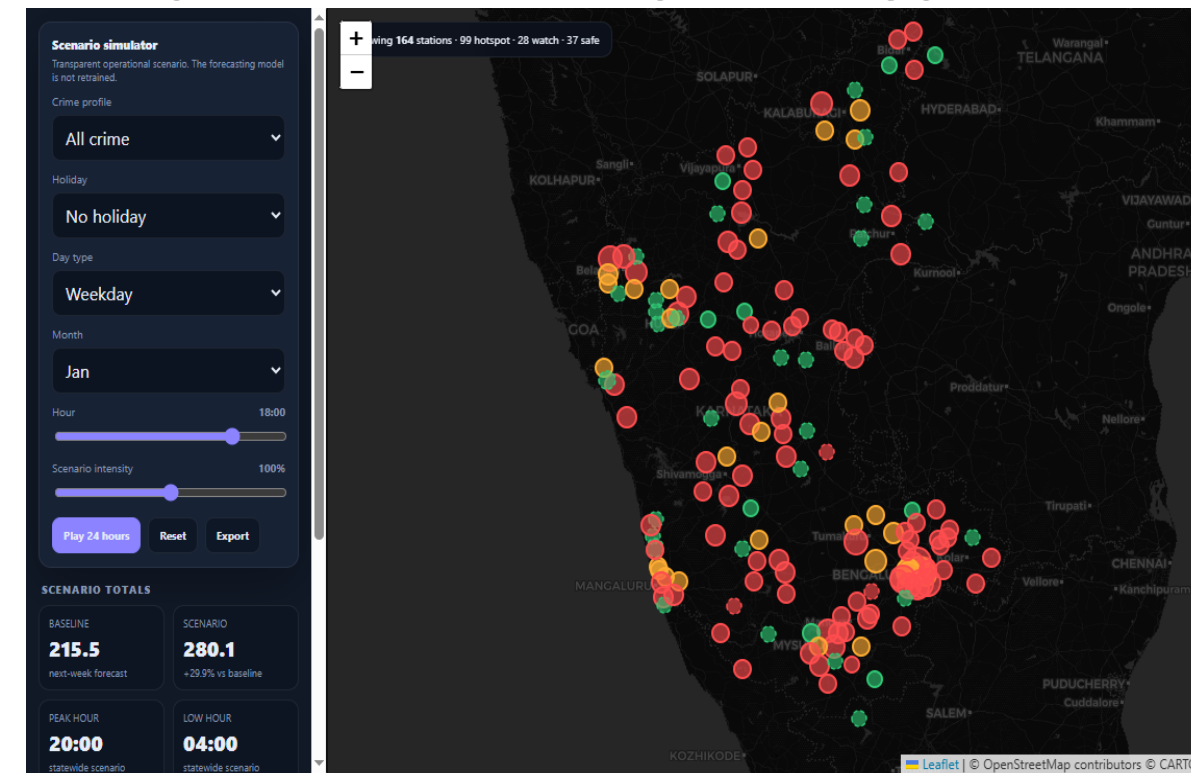


Prototype cost is low; production cost is driven by refresh frequency and governance

Prototype build	Low	Bundled data & outputs; static file serving
Hosted demonstration	Low	One containerized Catalyst AppSail instance
Operational pilot	Variable	Authoritative live data ingestion, Access Controls, Monitoring and Scheduled refresh.
Statewide production	Needs sizing	Statewide HA hosting, security & 24/7 SLA.



Snapshots of the prototype



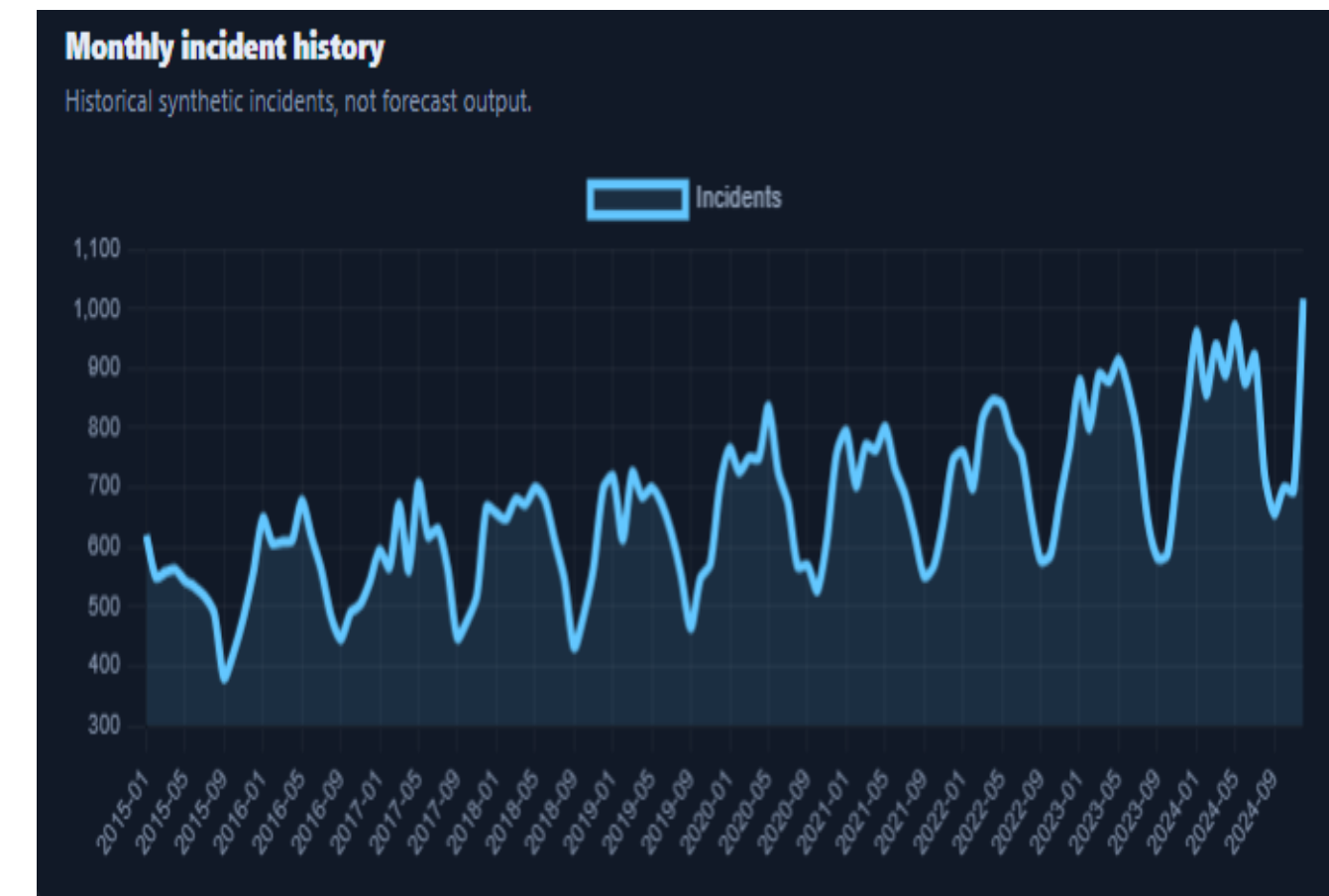
Prospective district PAI
Validation threshold and later untouched test efficiency.

District	Threshold	Val PAI	Test PAI	Zones
Bengaluru Urban	4.53	1.26	1.20	12
Belagavi	2.00	2.17	1.81	8
Mysuru	1.54	1.95	1.68	8
Dakshina Kannada	1.47	1.99	2.23	8
Tumakuru	3.27	2.09	1.71	5
Kalaburagi	2.28	3.07	2.44	5
Davanagere	1.91	2.16	1.04	5
Shivamogga	1.42	1.75	0.95	5
Ballari	1.13	1.34	0.99	5
Hubballi-Dharwad	1.64	2.21	2.57	8
Hassan	0.94	1.30	1.56	5
Mandya	1.04	1.90	1.21	5
Kolar	0.89	1.51	1.13	5
Vijayapura	0.97	1.35	1.15	5

MAP + SCENARIO SIMULATOR

FILTERED CYBER NETWORK + ROUTE MAP

DISTRICT WISE PREDICTIVE ACCURACY INDEX.



Trend Analysis :

- Monthly incident history
- Month Crime rate index
- Weekday Crime rate index

Prototype Performance Benchmarking

COUNT FORECAST BENCHMARK (MAE)



SERIOUS RISK & PAI METRICS

Average Precision
0.489

Serious ROC-AUC
0.707

Median District PAI
1.319

AUDIT & NETWORK BENCHMARKS

FAIRNESS AUDIT

- 1.40% arrest rate spread (9 groups)
- 0 audit warnings configured

CYBER NETWORK OPTIMIZATION

- 76% visible node clutter reduction
- 119 repeat accused & 559 links



Submission links

01 GitHub public repository

<https://github.com/bhaktictak/ksp-crime-intelligence>

02 Demo video (3 minutes)

03 Deployed prototype

<https://kspcrimeintelligence-50044296851.development.catalystappsail.in/>



FUTURE DEVELOPMENT: FROM SYNTHETIC VALIDATION TO GOVERNED OPERATIONAL EVIDENCE

NOW

Harden Demo

01

- Automated CI/CD
- Offline map caching
- Signed release (checksums (SHA256)
- , • Accessibility (WCAG).

PILOT

Authoritative Data

02

- Official KSP GIS boundary integration
- Secure CAD pipelines
- RBAC & Audit logging
- Scheduled retrain

SCALE

Prove operational value

03

- Prospective field evaluation,
- Model drift monitoring,
- Human review



RESPONSIBLE - DEPLOYMENT GUARDRAILS & GOVERNANCE

Decision support for synthetic engineering - NOT an automated enforcement system

SUPPORTED USES

- Reproduce & test pipeline code
- Benchmark vs. seasonal baselines
- Explore "What-If" scenarios
- Review Spatial, Network and Fairness Diagnostics

NOT SUPPORTED

- Treat labels as dispatch orders
- Claim legal fairness/causal bias
- Interpret routes as exact GPS
- Deploy without official boundaries

Descriptive diagnostics based on 80k synthetic FIRs



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THANK YOU